

OMBR - Open Modular Brick Robotics

Executive summary - open infrastructure for programmable brick automation

- **Version:** 0.1 proposal for public review
 - **Date:** 13 July 2026
 - **Audience:** Founding engineering, manufacturing, legal, education, and open-source partners
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The opportunity

Programmable brick systems have demonstrated the educational and engineering value of modular controllers, motors, sensors, and construction elements, but today's offerings remain fragmented. Brick-integrated products are generally closed at the electronics, interface, or lifecycle level. Maker platforms can be open and highly programmable but do not fit brick and studless-beam constructions. The products reviewed by OMBR do not document the full combination of Linux-class compute, Wi-Fi and Bluetooth, brick-compatible mechanics, open controller and peripheral designs, complete electrical contracts, and a synchronized CAD and simulation workflow.

The result is avoidable lock-in: undocumented connectors, unavailable parts, vendor-specific applications, incomplete engineering data, and systems that cannot be independently reproduced, repaired, extended, or simulated.

Executive proposition

OMBR defines a vendor-neutral system for modular brick automation. "Robotics" is historical shorthand: the same system covers robots, vehicles, conveyors, cranes, plotters, test rigs, instruments, kinetic installations, and other programmable machines. A creation moves through one continuous workflow: design in CAD, validate mechanics and wiring, simulate, program, deploy, operate, and synchronize measured state back to its digital twin.

The platform is local-first, has no mandatory cloud dependency, supports ordinary programming languages, and lets multiple manufacturers and community projects create interoperable components.

One system, three synchronized layers

- **Physical ecosystem:** A Raspberry Pi Zero-class controller, protected native ports, power sources, cables, motors, position actuators, lights, sensors, adapters, brick-grid structures, gears, transmissions, and test fixtures.
- **Open control plane:** Self-describing capabilities, complete electronic interface contracts, deterministic safety supervision, local APIs, portable Python and .NET SDKs, a headless service and CLI, a VS Code Workbench, and interchangeable standalone clients.
- **Digital engineering:** Editable parametric CAD, assemblies, joints and transmissions, collision and load information, simulation models, calibration, telemetry, recording, replay, and an offline-first digital twin using the same identities and limits as physical hardware.

Design principles

- **A whole ecosystem, not another hub:** The specification includes the controller, motors, sensors, lights, cables, mechanics, software, CAD, simulation, test evidence, repair, and conformance.
- **Complete interfaces:** Every electrical boundary declares its connector, pinout, protocol, voltage limits, minimum, continuous, peak, inrush, and fault currents, grounding, sequencing, cable constraints, protection, fault behavior, and evidence.
- **Honest openness:** Project-controlled PCB, firmware, CAD, fixtures, protocols, and tests are open. Proprietary silicon and commodity parts are identified as documented, replaceable COTS dependencies, not mislabeled as open hardware.
- **Physical and digital parity:** The same capability model drives a real motor or sensor and its simulated counterpart. The twin is an engineering record, not a disconnected visualization.

Where it applies

The initial audience spans makers, education, research, model engineering, product prototyping, laboratory automation, and non-safety-related stationary machines. Version 0.1 is not a safety-critical, road-going, medical, aviation, or production-machine controller.

Reference controller

The first controller combines a replaceable Raspberry Pi Zero 2 W for Linux, Wi-Fi, Bluetooth, programming, and digital-twin services with a separate real-time MCU, independent watchdog, and hardware output-disable path. Four protected native ports support motors and smart peripherals. Linux provides programmability and networking; the independent controller owns deadline enforcement, fault handling, and rapid output neutralization.

Implementation strategy

The shortest implementation path reuses documented commercial components inside an original open carrier and enclosure. Current screen candidates include RP2040 safety control, dedicated motor drivers and power protection, CAN FD data transport, qualified storage, and a provisional commodity connector. Every choice remains subject to bench, thermal, EMC, lifecycle, security, and cost qualification.

OMBR does not plan production moulds. It will maintain a bounded, source-traceable engineering catalog, use independently measured compatibility geometry where lawful, and publish original parametric parts for accessible processes such as FDM, resin printing, CNC, or documented service bureaus.

Commercial reality

The controller-only affordability objective is **EUR 49.99 including German VAT**, excluding the power supply, battery, charger, external cables, motors, sensors, lights, and construction elements. Its status is deliberately **target-unproven**.

The current pre-quote estimate for a complete four-port controller at 1,000 units is **EUR 39-57 COGS** before engineering, certification, legal review, long-term support, fulfilment, and channel costs. A more realistic initial direct-sale range is approximately **EUR 65-95 including VAT**. Reaching the target requires higher volume, connector and BOM cost reduction, manufacturing partners, measured yield and test time, and carefully amortized compliance cost. Safety, test evidence, repairability, and open source are not eligible cost cuts.

Open and lawful by design

OMBR is independent and is not affiliated with, authorized by, or endorsed by the LEGO Group. OMBR is a working technical identifier pending trademark clearance. Compatibility work preserves only necessary functional geometry, uses independent metrology, and creates original nonfunctional appearance. It does not copy logos, ornamental housings, or restricted CAD catalogs.

Open licensing does not by itself establish patent freedom to operate. The selected hub, connector, motor, sensor, gear, protocol, and software revisions must pass territory-specific trademark, patent, design, licensing, product classification, cybersecurity, and regulatory gates before commercial release. Patent-aware contribution terms, provenance records, public decision logs, and conflict disclosures are part of the platform rather than launch afterthoughts.

Delivery path

- **Phase 0 - prove the contracts:** Finalize schemas and governance; validate the Pi-to-safety-controller path, protected single-port bench hardware, connector candidates, simulator, and safety state machine.
- **Phase 1 - release Hub R0:** Publish an open four-port developer carrier, enclosure, firmware, fixtures, SDKs, VS Code integration, and quote-backed price book.
- **Phase 2 - establish the ecosystem:** Deliver open motor, position motor, RGBW light, distance and reflectance or force sensors, cables, and adapters with measured simulation models.
- **Phase 3 and 1.0 - close the physical-digital loop:** Release fit metrology, parametric mechanics and gears, the complete Workbench and live twin, then qualify a production connector, independent implementations, conformance tests, legal dispositions, and applicable product-compliance evidence.

Partnership opportunity

OMBR is seeking sponsors for prototype hardware, metrology, EMC and thermal testing, legal review, and open infrastructure; electronics partners for power, motor, safety, connector, RF, and DFM/DFT engineering; manufacturing partners for sourcing, pilot builds, test automation, and lifecycle planning; mechanical contributors for measurements, original CAD, gears, and fixtures; software contributors for schemas, SDKs, VS Code, .NET, simulation, digital twins, and visual programming; and educators and makers to define representative creations and reproducible acceptance tests.

Executive decision

OMBR is technically achievable with available components, but commercial credibility depends on disciplined validation, cost-down engineering, and rights and compliance work. The immediate decision is whether to fund and staff Phase 0 as a public, evidence-driven reference program. Success would create a durable common layer for programmable brick automation: open enough to inspect and reproduce, practical enough to build, and rigorous enough for independent implementations to work together.

This summary is informative. The normative requirements, evidence status, and current limitations are defined by the OMBR 0.1 specification, implementation catalog, and legal/IP register.